

Master in
Computer Vision
Barcelona

C2 – Optimization Techniques for Computer Vision
The Mumford–Shah functional and Segmentation Techniques

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Contact

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Segmentation

Segmentation is the task of **partitioning** an image into regions that can be distinguished by some **local characteristic** (brightness, color, texture) or by their semantics, while satisfying **regularity constraints** on the region boundaries.



Source: Jégou *et al.*, CVPR Workshops 2017 (arXiv:1611.09326)

Segmentation

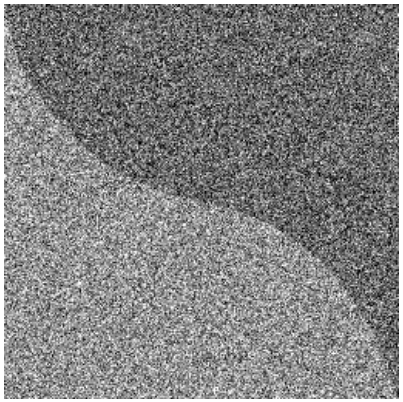
Examples of practical applications:

- **Medical imaging**: organs and lesions in MRI, CT or X-ray scans
- **Autonomous vehicles**: roads, pedestrians, vehicles, obstacles
- **Agriculture**: crops and soil in drone or satellite imagery
- **Cartography**: land use, vegetation, water bodies, urban change
- **Quality control**: defects on assembly lines (e.g., cracks in metal parts)
- **Security and surveillance**: anomaly detection in video streams
- **Entertainment**: object isolation/removal in photo and video editing, green screen, augmented reality

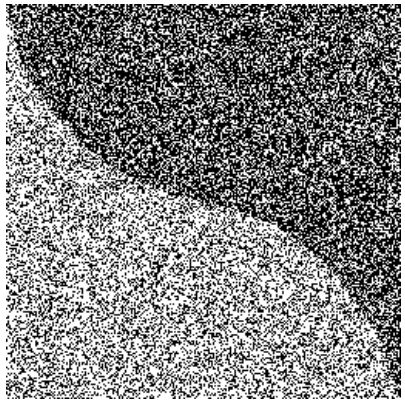
Why not simply threshold?

Two regions with different mean brightness, buried in strong noise (Lab 3):

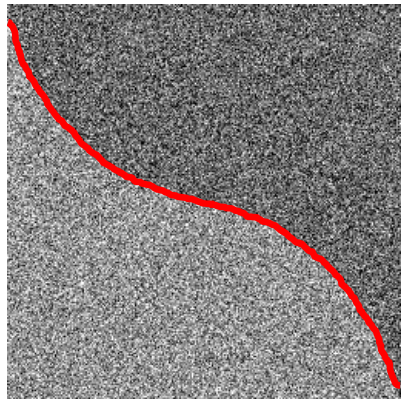
noisy image



Otsu threshold: 65 %



Chan–Vese: 99.7 %



- Thresholding decides **pixel by pixel**: the two histograms overlap, so many pixels fall on the wrong side.
- A variational model decides **region by region**: a boundary has a cost (its length), so isolated noisy pixels are not worth a boundary.
- No edge is visible anywhere, yet the boundary is found: "active contours **without edges**".

Conventional methods

Classical (non-learned) approaches to brightness-based segmentation:

- **Thresholding** and **clustering** (Otsu, k-means, Lloyd–Max): pixel-wise decisions
- **Edge-based**: edge detectors, active contours driven by the image gradient
- **Region-based variational models**: Mumford–Shah, Chan–Vese (this lecture)
- **Graph-based**: graph cuts, random walker

Variational models write segmentation as the minimization of an **energy**: each term encodes one assumption (regions are homogeneous, boundaries are short, . . .).

Variational models in the deep learning era

Strengths:

- **no training data**: every term of the energy is an explicit modelling assumption
- **interpretable** parameters (length, area, data fit), predictable failure modes
- **computationally light**; work on any modality and resolution
- still used **inside** deep pipelines, e.g. Mumford–Shah and active-contour **loss functions** to train networks with few labels

Drawbacks:

- **parameter tuning**
- **non-convex** energies: local minima, dependence on the initialization
- model mismatch: textures, illumination gradients, more than two regions, semantics

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Edge-based versus region-based

Segmentation is about determining the **boundary curves** C that delimit the regions.

Edge-based: geodesic active contours (Caselles, Kimmel, Sapiro, 1997). Minimize a weighted length

$$E(C) = \int_0^1 g\big(|\nabla (G_\sigma * f)(C(s))|\big)\,|C'(s)|\,\mathrm{d}s, \qquad g(r) = \frac{1}{1+r^2},$$

so that the curve is attracted by strong gradients (where g is small). Drawbacks:

- needs **edges**: fails with noise, blur, weak or missing edges; the curve can leak through gaps
- explicit parametric snakes (Kass, Witkin, Terzopoulos, 1988) cannot change topology (level sets can)

Region-based: model what is **inside** the regions, not their edges (Mumford–Shah 1989, Chan–Vese 2001). The boundary is wherever the model of the regions changes.

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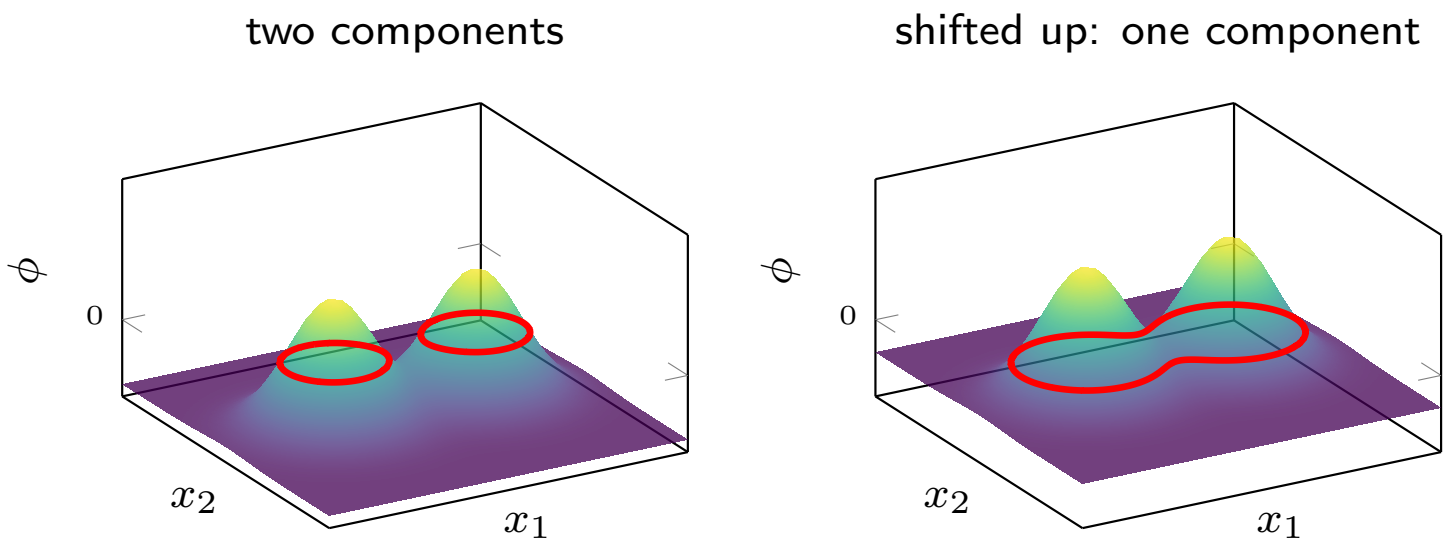
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Level set method

Two classes of curve representations:

- **Explicit**: $C = \{(x(t), y(t)) : t \in \mathcal{T}\}$ (snakes, parametric active contours)
- **Implicit**: C is the zero level set of an **embedding function** ϕ:
  $C = \{x : \phi(x) = 0\}$, inside $= \{\phi > 0\}$



Topology changes (merging, splitting) happen **automatically**: ϕ evolves smoothly, its zero level set (red) changes topology.

Curve evolution

Iterative methods design curves $C(t)$, parametrized by an artificial time t, that decrease an energy $E(C)$ as $t \to \infty$.

Level set equation. Convention: $\phi > 0$ inside, so the outward normal is $n = -\nabla\phi/|\nabla\phi|$. A point $x(t)$ of the curve stays on the zero level set:

$$\phi(x(t), t) = 0 \;\Rightarrow\; \frac{\partial \phi}{\partial t} + \nabla\phi \cdot x'(t) = 0.$$

If the curve moves with speed v_{n} along the outward normal, $x'(t) = v_{\mathrm{n}}\, n$, hence

$$\frac{\partial \phi}{\partial t} = v_{\mathrm{n}} |\nabla\phi|.$$

The embedding function is not unique. A convenient choice is the **signed distance function**

$$\phi(x, t) = \pm \operatorname{dist}(x, C) \qquad (+ \text{ inside}, - \text{ outside}),$$

which satisfies $|\nabla\phi| = 1$ almost everywhere.

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The Mumford–Shah functional

Mumford and Shah (1989): approximate $f : \Omega \subset \mathbb{R}^d \to \mathbb{R}$ by a function u that is smooth **except** on a closed discontinuity set $K \subset \Omega$:

$$E(u, K) = \underbrace{\int_\Omega (f-u)^2 \, \mathrm{d}x}_{\text{fidelity}} + \alpha \underbrace{\int_{\Omega \setminus K} \|\nabla u\|^2 \, \mathrm{d}x}_{\text{smooth away from } K} + \mu \underbrace{\mathcal{H}^{d-1}(K)}_{\text{size of } K}$$

$\mathcal{H}^{d-1}$ is the $(d-1)$-dimensional **Hausdorff measure**: the "size" of a set in the dimension it naturally has, even if it is not a smooth curve (junctions, cracks, isolated points).

- image ($d = 2$): $\mathcal{H}^1(K)$ = **length** of the edge curves
- signal ($d = 1$): $\mathcal{H}^0(K)$ = **number of jump points** (counting measure)
- ($\mathcal{H}^d$ is the usual volume: area in 2D, length in 1D)

For the curious: $\mathcal{H}^s(K) = \lim_{\delta \to 0} \ \inf\{\sum_i \omega_s (\frac{\operatorname{diam} U_i}{2})^s : K \subset \bigcup_i U_i, \ \operatorname{diam} U_i < \delta\}$: cover K by small sets and add up their diameters raised to the power s.

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The Mumford–Shah functional: properties

$$E(u, K) = \int_{\Omega} (f - u)^2 \, \mathrm{d}x + \alpha \int_{\Omega \setminus K} \|\nabla u\|^2 \, \mathrm{d}x + \mu \, \mathcal{H}^{d-1}(K)$$

- The minimizer is a **piecewise smooth** approximation of f; K is an edge map.
- Three competing terms: fidelity ($u \approx f$), smoothness away from K (scale $\sqrt{\alpha}$), economy of edges (weight μ).
- Geometry of optimal edge sets: **triple junctions** at 120°, edges meet the image border at 90°, crack tips may occur.
- Difficulties: two unknowns of different nature (a function and a set); the energy is **non-convex**.
- Practical approximations: Ambrosio–Tortorelli (an edge-indicator function), level sets, graph cuts.

To understand how the three terms interact, look at the simplest case: $d = 1$, where $\mathcal{H}^0$ just **counts jumps**.

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A 1D toy example: the weak string

Sampled signal $f_1, \ldots, f_N$; K = set of positions k where u may jump between k and $k+1$:

$$E(u, K) = \sum_{k=1}^{N} (f_k - u_k)^2 + \alpha \sum_{k \notin K} (u_{k+1} - u_k)^2 + \mu \, \#K.$$

(Blake and Zisserman, 1987: an elastic string that may **break**, at a cost μ per break.)

Exact global minimization by dynamic programming over the jump positions:

- between two jumps, u solves a small linear least-squares problem (tridiagonal system) with cost $c(l, r)$ for the segment $l+1, \ldots, r$;
- best energy of the first r samples: $B(r) = \min_{l<r} \big[B(l) + \mu + c(l, r)\big]$, $B(0) = -\mu$.

Special cases:

- $\mu \to \infty$: no jumps, $u =$ Tikhonov smoothing of f (blurs the edges)
- $\alpha \to \infty$: u piecewise **constant** (the "Potts model")

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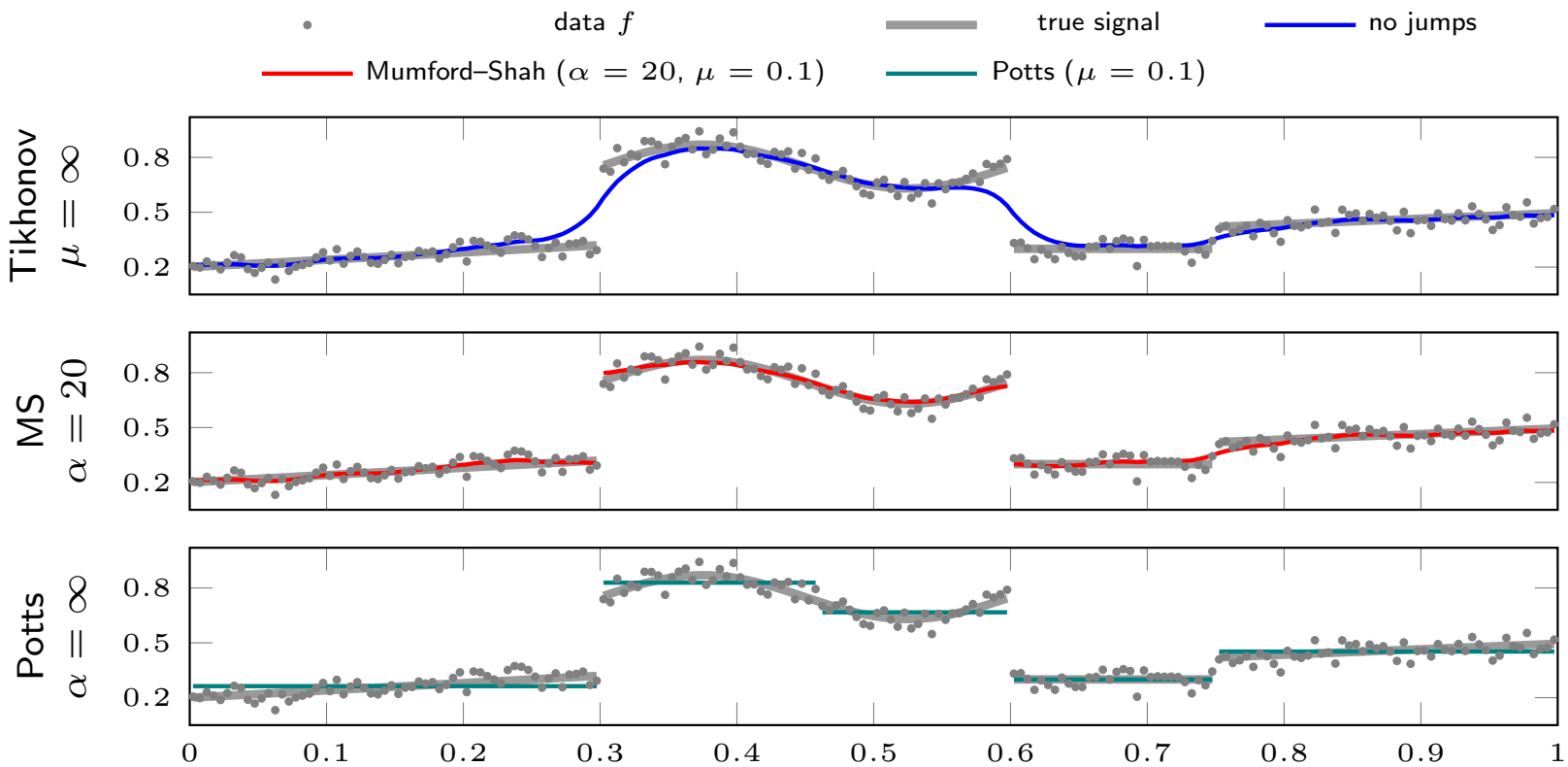
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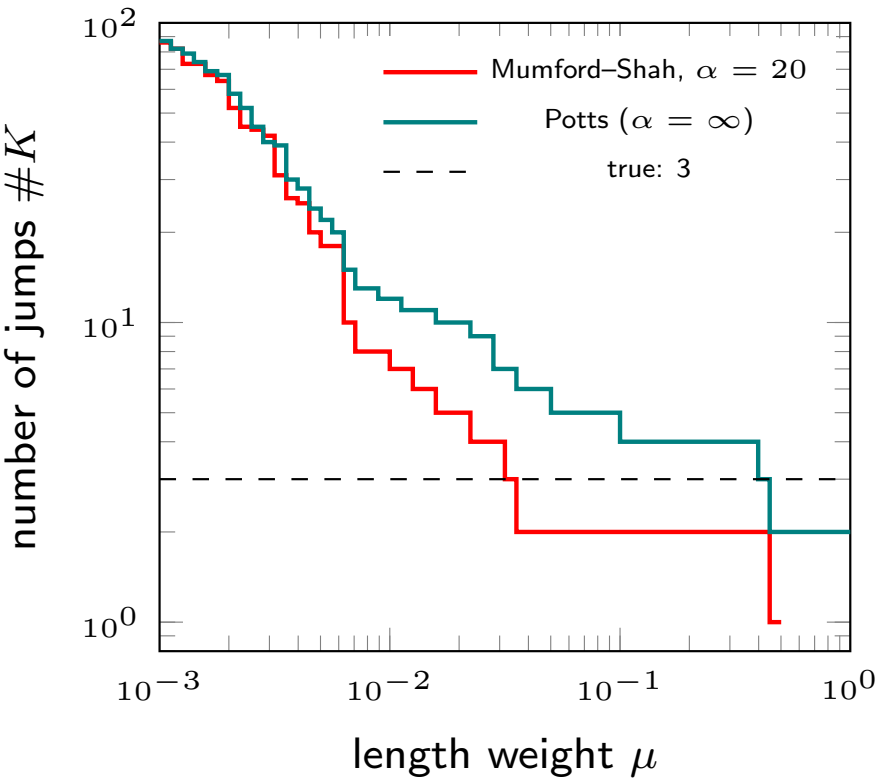
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1D toy example: three reconstructions



Tikhonov blurs every jump. Mumford–Shah keeps the two large jumps sharp, follows the ramps and the bump, and **smooths away** the small step at $x = 0.75$ (below the contrast threshold, next slide). The piecewise-constant model can only follow a variation with jumps: it splits the bump with an extra jump (**staircasing**) and keeps the small step.

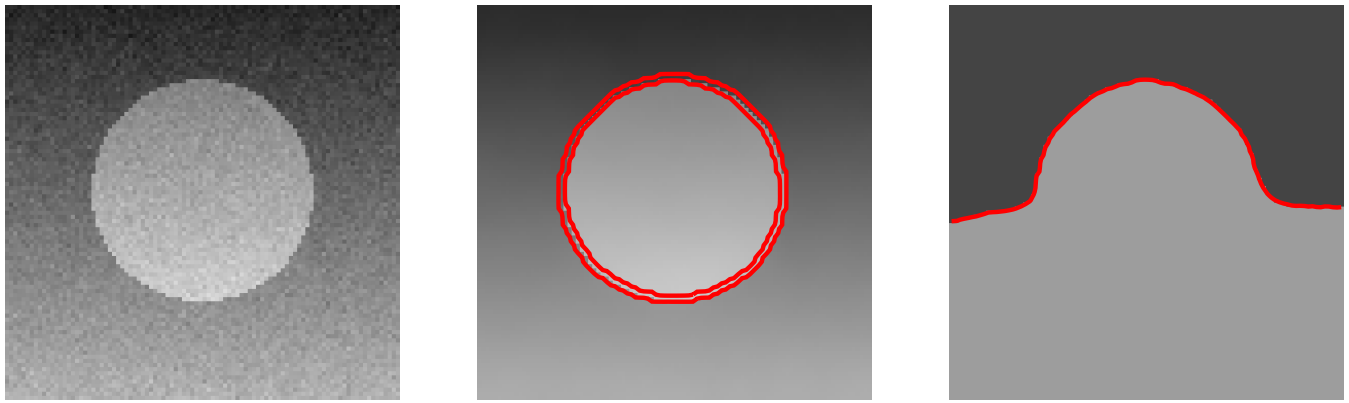
1D toy example: number of jumps versus μ



- Small μ: jumps are cheap, u follows the noise (many spurious jumps).
- Mumford–Shah: a long **plateau** with the two significant jumps (μ between ≈ 0.04 and 0.4); the third, small step lies below the contrast threshold.
- Potts: no plateau at the true structure; the ramps and the bump are approximated by extra jumps.
- Large μ: all jumps eventually disappear.

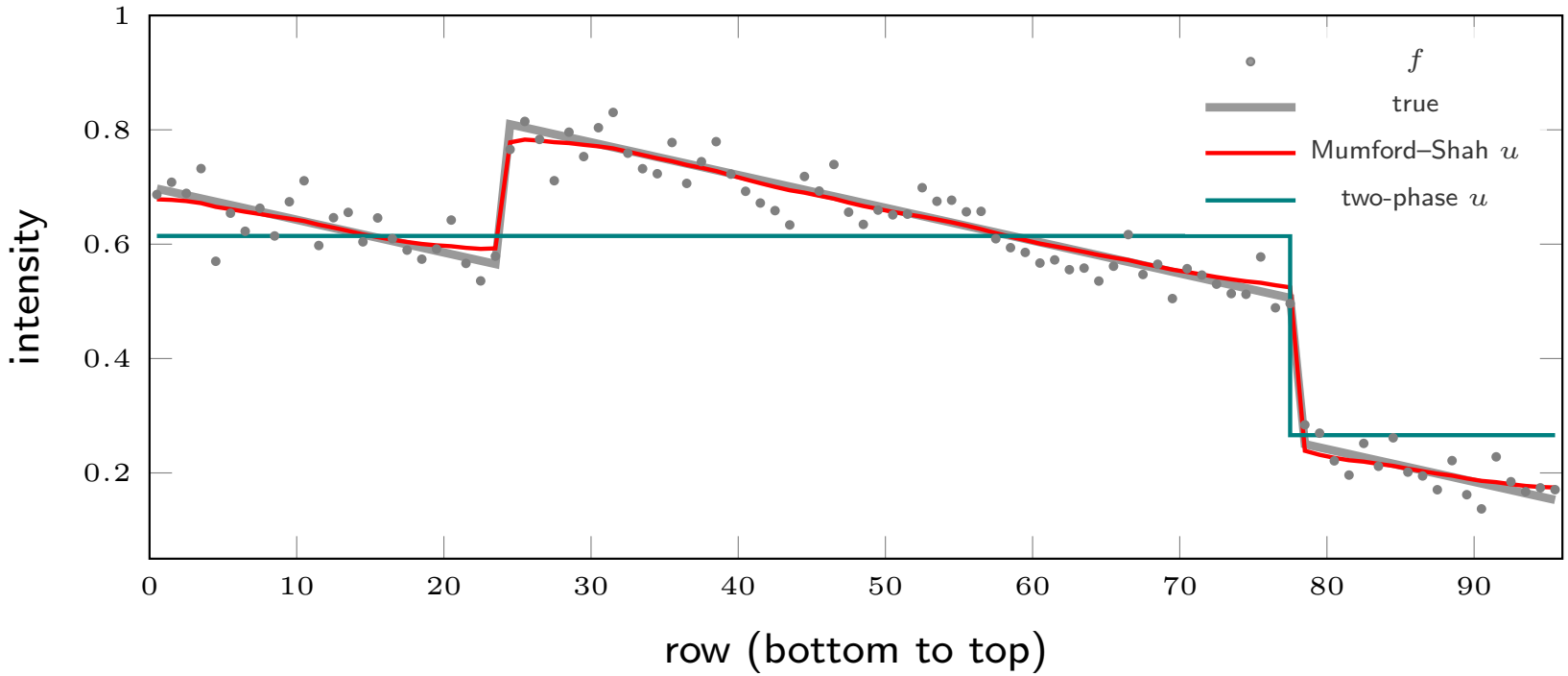
2D toy example: an object under an illumination gradient

data f | Mumford–Shah u, edges K | two-phase model



- The disk is brighter than its **local** surroundings everywhere, but its top is darker than the bottom of the background.
- Mumford–Shah (piecewise smooth) follows the ramp inside each region and puts the edges on the disk boundary (computed with the Ambrosio–Tortorelli approximation, $\alpha = 20$, $\mu = 0.06$).
- A two-constant model cannot: it cuts the image into a dark top and a bright bottom (61 % of pixels correct).

2D toy example: a vertical profile through the center



The Mumford–Shah reconstruction is a ramp with two sharp jumps; the two-phase model can only choose between two constant levels, and puts its single transition at the wrong place.

The piecewise constant Mumford–Shah model

Sending $\alpha \to \infty$ forces $\nabla u = 0$ on $\Omega \setminus K$: u is **piecewise constant**, and

$$E(u, K) = \int_\Omega (f - u)^2 \, \mathrm{d}x + \mu \, \mathcal{H}^1(K).$$

Denoting by $\Omega_1, \ldots, \Omega_n$ the connected components of $\Omega \setminus K$ and by c_i the value of u on Ω_i:

$$E(c_1, \ldots, c_n, K) = \sum_{i=1}^{n} \int_{\Omega_i} (f - c_i)^2 \, \mathrm{d}x + \mu \, \mathcal{H}^1(K).$$

- K consists of closed boundaries.
- For fixed K, the optimal c_i is the least-squares solution $c_i = \frac{\int_{\Omega_i} f \, \mathrm{d}x}{\int_{\Omega_i} \mathrm{d}x}$ (region average).
- These optimal c_i could be reinserted into the cost function (in principle).
- The dependence on Ω_i is tricky (it sits in the integration domains!).
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- The number of levels n is optimized automatically (some values can become equal).

The piecewise constant Mumford–Shah model

Sending $\alpha \to \infty$ forces $\nabla u = 0$ on $\Omega \setminus K$: u is **piecewise constant**, and

$$E(u, K) = \int_\Omega (f - u)^2 \, \mathrm{d}x + \mu \, \mathcal{H}^1(K).$$

Denoting by $\Omega_1, \ldots, \Omega_n$ the connected components of $\Omega \setminus K$ and by c_i the value of u on Ω_i:

$$E(c_1, \ldots, c_n, K) = \sum_{i=1}^{n} \int_{\Omega_i} (f - c_i)^2 \, \mathrm{d}x + \mu \, \mathcal{H}^1(K).$$

- K consists of closed boundaries.
- For fixed K, the optimal c_i is the least-squares solution $c_i = \frac{\int_{\Omega_i} f \, \mathrm{d}x}{\int_{\Omega_i} \mathrm{d}x}$ (region average).
- These optimal c_i could be reinserted into the cost function (in principle).
- The dependence on Ω_i is tricky (it sits in the integration domains!).
- The number of levels n is optimized automatically (some values can become equal).

The Chan–Vese model

Chan and Vese (2001): a level set method for the piecewise constant Mumford–Shah model with **two** regions, $\Omega_1 \subset \Omega$ ("inside") and $\Omega_2 = \Omega \setminus \Omega_1$ ("outside"):

$$E(c_1, c_2, \Omega_1) = \mu\, \ell(\partial\Omega_1) + \nu\, |\Omega_1| + \lambda_1 \int_{\Omega_1} (f - c_1)^2 \,\mathrm{d}x + \lambda_2 \int_{\Omega_2} (f - c_2)^2 \,\mathrm{d}x$$

(μ: length, ν: area, λ_1, λ_2: data weights; usually $\lambda_1 = \lambda_2 = 1$, $\nu = 0$.)

With the Heaviside function $H(t) = 1$ for $t \geq 0$, 0 for $t < 0$, the inside is $\{H(\phi) = 1\}$ and

$$\begin{aligned} E(c_1, c_2, \phi) = \mu \int_\Omega \|\nabla H(\phi)\| \,\mathrm{d}x + \nu \int_\Omega H(\phi) \,\mathrm{d}x \\ + \lambda_1 \int_\Omega (f - c_1)^2 H(\phi) \,\mathrm{d}x + \lambda_2 \int_\Omega (f - c_2)^2 \big(1 - H(\phi)\big) \,\mathrm{d}x. \end{aligned}$$

The first term is the **total variation** of the indicator $H(\phi)$, i.e. the perimeter of the inside region.

Regularized Heaviside and Dirac functions

Length of the boundary:

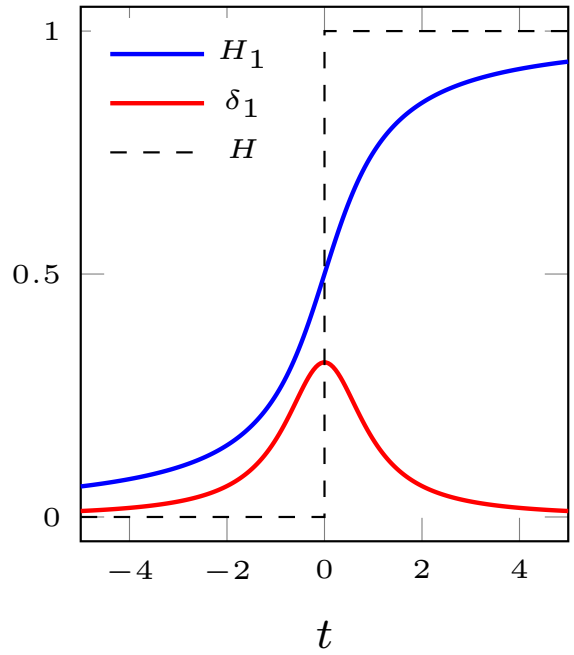
$$\ell(\partial\Omega_1) = \int_\Omega \|\nabla H(\phi)\| \, \mathrm{d}x$$
$$= \int_\Omega \delta(\phi)\|\nabla\phi\| \, \mathrm{d}x.$$

To differentiate with respect to ϕ, H and δ are replaced by smooth versions:

$$H_\epsilon(t) = \frac{1}{2}\left(1 + \frac{2}{\pi}\arctan\frac{t}{\epsilon}\right),$$
$$\delta_\epsilon(t) = H'_\epsilon(t) = \frac{\epsilon}{\pi(\epsilon^2 + t^2)},$$

with $\epsilon = 1$ (pixel units).

- δ_ϵ has **heavy tails**: points far from the curve still evolve (slowly), so new components can appear away from C.



Alternating minimization

Alternate between (c_1, c_2) and ϕ.

For fixed ϕ, the optimal c_1 and c_2 are the average brightness levels of the two regions:

$$c_1 = \frac{\int_\Omega f\, H(\phi)\, \mathrm{d}x}{\int_\Omega H(\phi)\, \mathrm{d}x}, \qquad c_2 = \frac{\int_\Omega f\, \big(1 - H(\phi)\big)\, \mathrm{d}x}{\int_\Omega \big(1 - H(\phi)\big)\, \mathrm{d}x}.$$

For fixed (c_1, c_2), ϕ follows the **gradient descent** (gradient flow)

$$\frac{\partial \phi}{\partial t} = \delta_\epsilon(\phi)\Big[\mu\, \mathrm{div}\Big(\frac{\nabla \phi}{\|\nabla \phi\|}\Big) - \nu - \lambda_1 (f - c_1)^2 + \lambda_2 (f - c_2)^2\Big] \quad \text{in } \Omega,$$

$$\frac{\delta_\epsilon(\phi)}{\|\nabla \phi\|}\frac{\partial \phi}{\partial n} = 0 \quad \text{on } \partial\Omega.$$

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